Pandas Practice Questions

1. Load and Understand Data

1. Read the CSV file using Pandas.
2. Display the first 5 rows.
3. Display the last 5 rows.
4. Display 10 random rows.
5. Find the number of rows and columns.
6. Display all column names.
7. Check the data types of all columns.
8. Generate summary information.
9. Generate statistical summary.
10. Check whether the dataset is empty or not.

2. Selecting Columns

11. Display only the `title` column.
12. Display `title` and `type`.
13. Display `title`, `country`, `release_year`, and `rating`.
14. Display the first 20 rows containing only `title` and `director`.

3. Basic Filtering — `loc` / `[ ]`

15. Find all Movies.
16. Find all TV Shows.
17. Find movies released after 2015.
18. Find titles released in 2020.
19. Find movies having rating `TV-MA`.
20. Find records where `country` is `United States`.
21. Find movies released before 2000.
22. Find titles where `release_year` is greater than 2010.
23. Display only `title` and `release_year` for movies released after 2018.

4. Missing Values & Duplicates

24. Find the number of missing values in every column.
25. Find the percentage of missing values in every column.
26. Find rows where `director` is missing.
27. Find rows where `country` is missing.
28. Find rows where `rating` is missing.
29. Count how many missing values are present in `duration`.
30. Replace missing values in `director` with `"Unknown"`.
31. Replace missing values in `country` with `"Unknown"`.
32. Replace missing values in `rating` with `"Not Rated"`.
33. Remove rows where `title` is missing.
34. Check whether duplicate rows exist.
35. Count the number of duplicate rows.
36. Remove duplicate rows.
37. Find duplicate values based only on `title`.

5. Sorting and Value Counts

38. Sort the dataset by `release_year` in ascending order.
39. Sort by `release_year` in descending order.
40. Find the 10 oldest titles.
41. Find the 10 newest titles.
42. Count the number of Movies and TV Shows using `value_counts()`.
43. Count titles by `rating`.
44. Find the top 10 most common ratings.
45. Count titles by `release_year`.
46. Find how many titles were released in each year.
47. Find the top 10 years having the highest number of releases.
48. Count titles by `country`.
49. Find the top 10 countries with the highest number of titles.

5. String Functions

50. Find titles containing the word `"Love"`.
51. Find titles containing `"the"` ignoring uppercase/lowercase.
52. Find titles starting with `"A"`.
53. Find titles ending with `"!"`.
54. Convert all titles to lowercase.
55. Convert all country names to uppercase.

56. Find the length of every title.
57. Find the title with the maximum number of characters.
58. Find the title with the minimum number of characters.
59. Create a new column called `title_length`.
60. Find titles having more than 30 characters.
61. Find directors whose names contain `"John"`.
62. Find records where `cast` contains `"Tom"`.
63. Count how many actors are listed in the `cast` column for each record.

[Hint: df["cast"].str.split(",").str.len()]

6. `apply()` and Creating Columns

64. Create a column `is_movie` that contains:
- `1` if type is Movie
- `0` otherwise
65. Create a column `content_age`:

2026 - release_year

66. Create a column `era`:
- Before 2000 → `"Old"`
- 2000–2010 → `"Medium"`
- After 2010 → `"Recent"`
67. Create a column `title_length` using `apply()`.
68. Create a column `duration_minutes` by extracting the numeric part from `duration`.

For example:

"90 min" → 90

"2 Seasons" → 2

69. Create a column `duration_type`:
- `"Minutes"` for Movies
- `"Seasons"` for TV Shows

7. `groupby()`

70. Find the number of titles for each `type`.
71. Find the number of titles released in each year.
72. Find the number of Movies released in each year.
73. Find the number of TV Shows released in each year.
74. Find the number of titles for each rating.
75. Find the number of titles for each country.
76. Find the average release year for each type.
77. Find the minimum release year for each type.
78. Find the maximum release year for each type.
79. Find the number of titles for each `country` and `type`.
80. Find the number of titles for each `rating` and `type`.

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8. `agg()` / Multiple Aggregations

81. For each `type`, find:

- Number of titles
- Minimum release year
- Maximum release year
- Average release year

Use:

groupby()

agg()

82. For each rating, find:

- Count
- Minimum release year
- Maximum release year

83. For each country, find:

- Number of titles
- Average release year

84. Find the number of titles and average release year for each `type` and `rating`.
85. Find the country having the highest number of Netflix titles.

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9. DateTime Functions

The `date_added` column contains values such as:

January 1, 2020

August 5, 2017

November 20, 2020

86. Convert `date_added` into datetime format.
87. Extract the year from `date_added`.
88. Extract the month from `date_added`.
89. Extract the month name.
90. Extract the day.
91. Create a new column `added_year`.
92. Find how many titles were added in each year.
93. Find how many titles were added in each month.
94. Find the year in which Netflix added the highest number of titles.
95. Find all titles added after 2019.

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10. Multiple Conditions

96. Find Movies released after 2015 with rating `TV-MA`.
97. Find TV Shows released after 2018.
98. Find Movies released between 2010 and 2020.
99. Find titles from the United States released after 2015.
100. Find titles where:

type = Movie

AND

release_year > 2015

AND

rating = TV-MA

101. Find titles that are either: Movie OR TV Show using `isin()`.
102. Find titles released between 2000 and 2010 using `between()`.
103. Find titles where country is either United States, India, or Japan.